

A simple method for estimating mutual diffusion coefficients of ionic liquids-water based on an optofluidic chip



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ABSTRACT

An optofluidic method based on a liquid-core and liquid-cladding optical waveguide is proposed for roughly estimating the extent of diffusive mixing between two fluids. The mutual diffusion coefficient was estimated from a charged coupled device (CCD) camera image of light-focusing in accordance with predictions of a three-dimensional model. To check the validity of our proposed method, mutual diffusion coefficients of aqueous solutions of ethylene glycol at different volume fractions were measured, firstly. Then, mutual diffusion coefficients of eight 1,3-methylimidazolium ionic liquids-water binary systems were estimated at 298.2 K. Our results show that the mutual diffusion coefficients, in order of decreasing values, are as follows: [Emim][N(CN)₂] > [Mmim][BF₄] > [Emim][BF₄] > [Bmim][NO₃] > [Bmim][N(CN)₂] > [Emim][ClO₄] > [Bmim][BF₄] > [Bmim][ClO₄], it is well correlated to molecular weights of the ionic liquids. Finally, mutual diffusion coefficients of three ionic liquids-water binary systems at temperatures ranging from 298.2 to 338.2 K were measured, which is sensitive to temperature and increases quickly with it. Our results show a moderate discrepancy, but this proposed optofluidic method allows one to perform diffusion measurements with a simple, costless and microlitre volume manner in highly confined spaces.

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1. Introduction

Room temperature ionic liquids (RTILs) have been promoted as the green and “designer” solvents of the next generation with many potential applications, as they have various unique properties (such as designability, low volatility, wide liquidus, wide liquid temperature range, and thermal stability) that can be custom-fitted and would have lesser degrading effects on the environment [1,2]. Physical properties of RTILs must be available for process design and development in their applications. One of the most important physical properties that determines the ability of ionic liquids to spread is their diffusion coefficient in solvent. It is relevant to many physicochemical processes, such as thermal separation processes, catalysis, electrochemistry, transport properties and mixtures with suitable solvents [3]. Therefore, understanding mass transport in liquids by mutual diffusion is an important topic where ionic liquids are used in these applications. In addition, RTILs could be used as a novel class of soft material for creating continuously

tunable optofluidic devices and controlling blend of light by diffusion, due to its miscibility with other solvents and good optical transparency in visible light, near-infrared and even in some other infrared regions [4]. Reliable diffusion coefficients data are important for designing microfluidic and optofluidic devices efficiently that rely on precise transport of molecules. Despite this, there were very few systematic data of transport properties of ionic liquids in solutions, even in water [3,5,6]. For those reasons the study of diffusion coefficients of ionic liquids is of particular interest.

Many different techniques such as the widely employed Taylor diffusive dispersion analysis [7], nuclear magnetic resonance spectrometry [8], molecular theories, flow field flow fractionation and light scattering [9] have been reported to assess material transport parameters in solutions. Only little experimental measurements are available even for (the most common) diffusion in aqueous media and most of measurement methods are intricate or inconvenient for routine use [10,11].

The current trend in analytical chemistry is toward the miniaturization of chemical assays in an effort to exploit the many advantages of microfluidics [12]. Recently, a T shaped microfluidic channel (T-sensor) [13] and shear-driven flow microfluidic method [14] have been employed to determine diffusion coefficients of large and small molecules. Technological advances in the fields

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Nomenclature

Symbols

C	concentration (mol L^{-1})
c	volumetric concentration (%)
D	diffusion coefficient (m^2/s)
U	velocity (m/s)
λ	wavelength (nm)
$[\eta]$	intrinsic viscosity
μ	dynamic viscosity (mPa s)
n	refractive index
T	temperature (K)

of microfluidics and photonics have led to the development of optofluidics, in which photonic and microfluidic architectures are integrated to synergistically provide enhanced function and performance. It has been used in a wide range of fields including design of photonic devices, optical manipulation and energy [15,16]. In particular, optofluidic is well-suited for biological/chemical detection and analysis in extremely small detection volumes [17].

This work proposed an optofluidic method for measuring mutual diffusion coefficient of two liquids based on a liquid-core and liquid-cladding (L^2) optical waveguide [18]. In such a L^2 optical waveguide, the mutual diffusion creates refractive index gradients in both transverse and longitudinal directions, which causes the light to repeatedly merge in the waveguide. The key idea of this work is that a simple method for estimating mutual diffusion coefficients by measuring the focus length of light in L^2 optofluidic waveguides. Here, a series of diffusion coefficients of RTILs-water binary systems were estimated at different temperatures by finding the best fit between the numerical simulations and the experimental data. The mutual diffusion coefficients of aqueous solutions of ethylene glycol (EG) at different volume fractions were measured to check the validity of our proposed method. Our method offers some important advantages for estimating the mutual diffusion coefficient. First of all, this L^2 optofluidic waveguide chip can be mass-fabricated, resulting in a low cost sensor. Secondly, this method only needs extremely small detection volumes (as little as $1 \mu\text{l}$) even in a continuous-flow format. Thirdly, the optofluidic chip has potential advantages to integrate all photonic devices and micro-pump onto an optofluidic chip for measuring mutual diffusion coefficients in highly confined spaces.

The remainder of this paper is structured as follows: Section 2 describes the experimental setup and working principles; Section 3 discusses the measurement results; Section 4 provides the conclusions.

2. Working principle and experimental setup

2.1. Experimental system

The schematic diagram of the experimental configuration is illustrated in Fig. S.2. The system consists of five parts: PDMS chip, syringe pump, temperature control, laser light source and detector.

The optofluidic chip (Fig. 1) channels were fabricated by the standard soft lithography processes using the most popular polymer, polydimethylsiloxane (PDMS), due to its advantages in optical transparency and good optical properties. The fabrication of the chip is described in supplementary material. The width of the core channel, two cladding channels and waveguide channel is $100 \mu\text{m}$, $200 \mu\text{m}$ and $300 \mu\text{m}$, respectively. The height of all of the channels used in this experiment is $62 \mu\text{m}$.

The syringe pump was used to drive the fluids since it permitted the volumetric flow rates of the cladding and core flows

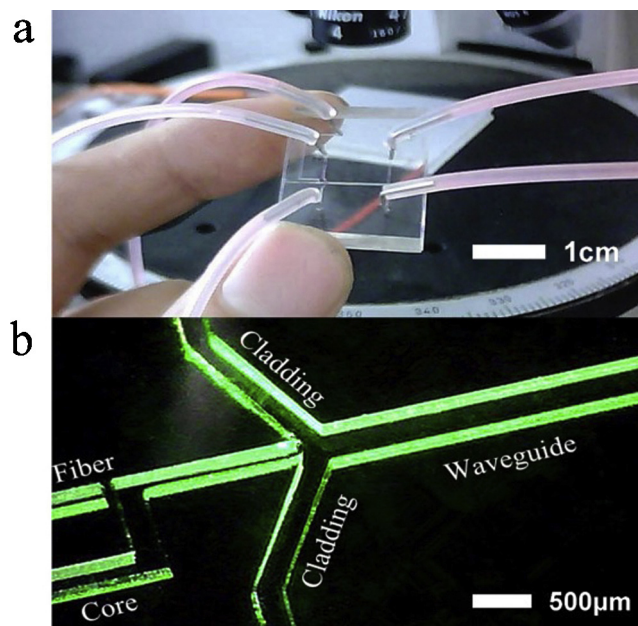


Fig. 1. (a) PDMS chip. (b) Microstructure image of the top layer with laser exposure before sealed.

to be precisely controlled. The two cladding flows and the core flows were stored in three 1 ml plastic syringes and driven through three positive displacement syringe pumps (LongerPump, LSP01-1A), respectively. The flow rate was fixed at the low end ($Q_{\text{cladding}} = 1 \mu\text{l/min}$ and $Q_{\text{core}} = 0.2 \mu\text{l/min}$) by the need to allow enough diffusion to occur. Note that, this experiment is very sensitive to flow rates and can be easily affected by tiny vibration in such low flow rates. To create a steady flow, the syringe pumps need to work at least 3 min before measurement. The two outlets were left open to the waste reservoir.

The optofluidic chip was mounted on an external Peltier element to control the temperature. This external Peltier facilitates both homogeneous temperature regulation within the whole microsystem, and linear temperature profiles with a high degree of accuracy. The virtue of PDMS is its relatively low thermal conductivity (0.15 W/mK typically), which allows efficient heat transfer from the source toward the liquid (minimizing energy losses) [19]. Meanwhile, the low flow speed (less than 1.34 mm/s) also allows efficient heat transfer from the source toward the liquid before the core stream and the two cladding streams flow into the waveguide. To keep the system at a constant temperature, the system was preheated for 5 min with Peltier before measurement.

Light from a free-space neodymium-doped yttrium aluminum garnet (Nd:YAG) laser ($\lambda = 532 \text{ nm}$, $P_{\text{CW}} = 50 \text{ mW}$) was coupled into the PDMS chip using a Multimode optical fiber (cladding diameter: $125 \mu\text{m}$, core diameter: $62.5 \mu\text{m}$) with a numerical aperture of 0.22. The fiber was inserted into a pre-fabricated microchannel with a width of $125 \mu\text{m}$ to work approximately as a light source. The optical path was perpendicular to the direction of flow of the fluids and pointed toward the center of the waveguide.

The experiments were conducted on a microscope (Nikon, XS-402), with the light that exits the L^2 waveguide was imaged by a charged coupled device (CCD) color camera mounted on the microscope. To facilitate observation and measurement of the ray trajectories, Rhodamine B (excitation: 540 nm , emission: 585 nm) with a concentration of $1 \times 10^{-6} \text{ mol L}^{-1}$ was added into the core liquid. The fluorescence images were then collected after passing through a long-wavelength pass optical filter (cutoff wavelength $\lambda_c = 560 \text{ nm}$).

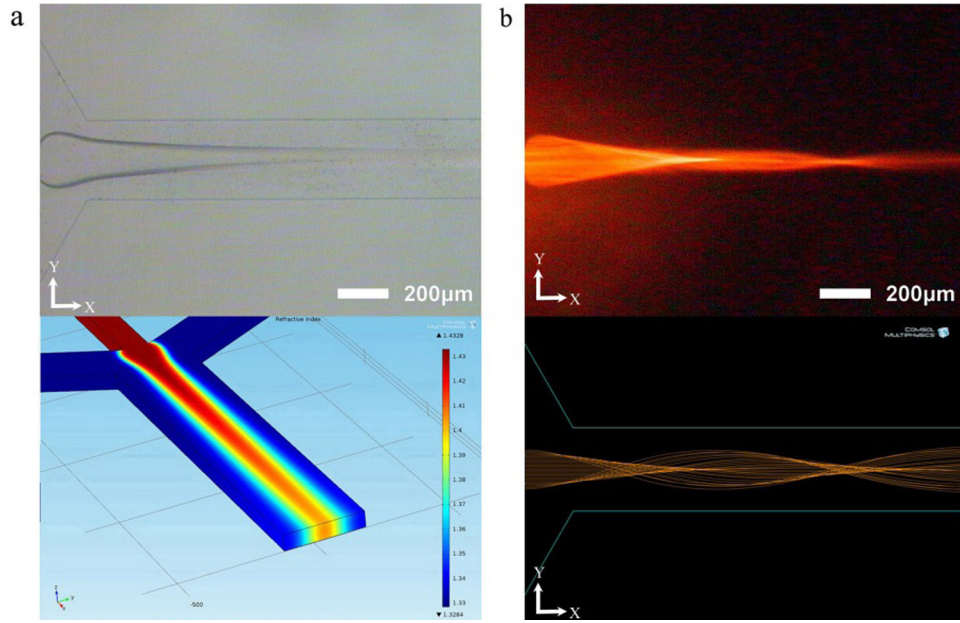


Fig. 2. (a) Refractive index profile in the L^2 waveguide. (b) The optical micrograph of light distribution in the waveguide with the refractive index profile shown in (a) (Top, experiment; Bottom, three-dimensional simulation).

2.2. Working principle

As shown in Fig. 2, the L^2 optofluidic waveguide consists of three flow streams – a core stream sandwiched by two cladding streams – to flow laminar in the microfluidic channel at low Reynolds number condition. The central flow stream has a higher refractive index than the side-flow streams. In such a L^2 optofluidic waveguide, a gradient index refractive profile can be formed in both the longitudinal and transverse directions (that is, x and y directions) due to mutual diffusion of the core and the cladding liquids, as shown in Fig. 2(a). The light undergoes a focusing effect when it propagates along x directions in such spatial distribution of the refractive index, as shown in Fig. 2(b).

The focus length associates with the mutual diffusion coefficient of the core stream and the cladding streams. Based on the above idea, we can estimate the mutual diffusion coefficient through experimental measure the first focus length and finding the best fit between the experimental results and the predicted results of numerical simulation.

2.3. Quantification using numerical simulations

The optofluidic waveguides is prominently governed by the diffusion and convection process between the core liquid and the cladding liquid. The convective-diffusive transport can be expressed as [20]

$$\frac{\partial c}{\partial t} = D \nabla^2 C - U \nabla C \quad (1)$$

where C is the concentration of solute, t is the time, D is the diffusion coefficient and U is the velocity. For a steady-state flow the concentration does not vary with time, and thus, the term on the left side $\partial c / \partial t = 0$ in Eq. (1). The first term on the right side corresponds to the diffusion process while the second term describes the convection process.

To determine the velocity profile of Eq. (1), we must solve Navier–Stokes equations [21]. As is well known, the dynamic viscosity depends on the concentration and the viscosity has a large influence on the liquid's velocity profile, especially in such a L^2 waveguide system, as shown in Fig. S.3. Dynamic viscosity can be

interpreted approximately with concentration by the Ward model [22]

$$\mu = \mu_{\text{cladding}} (1 + [\eta]c + ([\eta]c)^2 + ([\eta]c)^3) \quad (2)$$

where $[\eta] = \lim_{c \rightarrow 0} (\mu - \mu_0) / \mu_0 c$ is intrinsic viscosity, a dimensionless number and a measure of a solute's contribution to the viscosity of a solution, it depends on the solute (core stream), solvent (cladding stream) and temperature and can be determined by single-point method [23]. μ_0 is the viscosity in the absence of the solute, c is the volumetric concentration of solute (%). μ_{cladding} is the dynamic viscosity of solvent.

Next, the refractive index profile $n(x, y)$ can be derived from the concentration profile of the two liquid components

$$n(x, y) = \sum_{m=1}^2 c_m(x, y) n_m \quad (3)$$

Three types of numerical simulations were conducted by using the finite element technique. Firstly, three-dimensional hydrodynamic and mass transports were done with commercial software COMSOL Multiphysics 4.3a for calculating the concentration distribution, where the Navier–Stokes and the convection–diffusion equations were solved simultaneously. Based on the data of the concentration distribution, we obtained the refractive index distribution by Eq. (3) and then simulated the trajectories of the light beams with the geometrical optics method.

After collecting the experimental data from the CCD camera, the focus length of the first section was measured using a MATLAB program by finding max fluorescence intensity along the central line (x axis) of the waveguide, where an average fluorescence intensity from 5 lines of pixels centered at x axis was used. Finally, the diffusion coefficient was determined by finding the best fit between the experimental results and the predictions of the three-dimensional numerical simulation. This procedure needs to run the numerical simulation 3–5 times, which is the most time-consuming step for the whole measurement and need 10–15 min.

Table 1
The details of the chemicals used.

Chemical name	Source	Purification method	Purity (mass%)	Analysis method
[Mmim][BF ₄]	Synthesized in our lab: Lanzhou Institute of Chemical Physics, Chinese Academy of Sciences., China	Dissolving IL in dichloromethane; removing salts by washing IL 5 times with deionized water; distilling	>0.99	¹ H NMR ^a , Br contents ^b , Water content ^c
[Emim][BF ₄]			>0.99	
[Bmim][BF ₄]			>0.99	
[Emim][N(CN) ₂]			>0.99	
[Bmim][N(CN) ₂]			>0.99	
[Emim][ClO ₄]			>0.99	
[Bmim][ClO ₄]			>0.99	
[Bmim][NO ₃]		removing water by drying in vacuum.	>0.99	
Ethylene glycol	Tianjin Chemical Reagents Ltd., China	None	>0.995	Analyzed by a HP 6890/5973 GC–MS

^a The data of ¹H NMR was listed in supplementary material.

^b The Br contents of all the ILs are lower than 0.1 wt.%.

^c The water contents of all the ILs are lower than 300 ppm.

2.4. Room temperature ionic liquids

Ionic liquids (ILs) based on the 1,3-methylimidazolium cation have generated a great deal of interest over the past few years as solvents in many different areas of chemistry due to their ease of preparation. In our experiment, eight 1,3-methylimidazolium ILs *e.g.*, [Mmim][BF₄], [Emim][BF₄], [Bmim][BF₄], [Emim][N(CN)₂], [Bmim][N(CN)₂], [Emim][ClO₄], [Bmim][ClO₄] and [Bmim][NO₃] were synthesized according to the literature method [24–26] and were chosen as core streams in subsequent studies. The characterization of ionic liquids was assessed by ¹H Nuclear magnetic resonance (NMR) spectra. For each IL, the ¹H NMR spectrum contains peaks corresponding to their cation and indicates no residual reactants. The Br contents and water contents of all the ILs are lower than 0.1 wt.% and 300 ppm, respectively. All the ILs are transparent and colorless. The source, purification method and purity of the chemicals are reported in Table 1. De-ionized water was chosen as the cladding stream.

3. Results and discussion

We first checked the validity of the proposed method in this work by measuring mutual diffusion coefficients of aqueous solutions of EG at different volume fractions and compared the measurements against the data in available literature. Then, we extended to two experiments, one estimated the mutual diffusion coefficients of eight RTILs for 50% volume fraction at the temperature of 298.2 K, another estimated the diffusion coefficients of three RTILs for 50% volume fraction at temperatures ranging from 298.2

to 338.2 K. All of the mutual diffusion coefficient measurements of reported are average of three replicate runs. The flow rate was fixed $Q_{\text{cladding}} = 1 \mu\text{l/min}$ and $Q_{\text{core}} = 0.2 \mu\text{l/min}$ throughout the paper unless otherwise specified.

As is well known, the diffusion coefficient depends on the concentration of the solution for aqueous binary system [6]. To check the validity of the method used in this work, mutual diffusion coefficients of aqueous solutions of EG at different volume fractions were measured at 298.2 K and atmospheric pressure by using our optofluidic method described in Section 2. Fig. 3 show normalized fluorescence intensity profiles of three different mole fractions of EG (0.0698, 0.242 and 0.4119) along the central line (*x* axis) of the waveguide. It should be noted that we determined the focus length with the smoothed curve that is the fitted curve of experimental data as shown in Figs. 3 and 5, which might give rise to possible tiny errors. The experimental results were listed in Table 2. It is observed that the diffusion coefficient decreases with the increase of volume fraction of EG. The comparison, shown in Fig. 4, suggests that the estimated data nearly agree with the data reported in literature. The smoothed line was determined from literature [11,27] in Fig. 4. Compared with the literature value, our experimental results differed from accepted results by an average of 15%, due to some factors such as molecule-wall interactions and the effect of rough surface of microchannel may not be considered in the simulation. However, this method is valid and still can be used to roughly estimate the mutual diffusion coefficient of two liquids.

Next, we extend this method to estimate the mutual diffusion coefficients of RTILs–water binary systems. We choose pure RTILs as core stream and deionized water as cladding stream. The

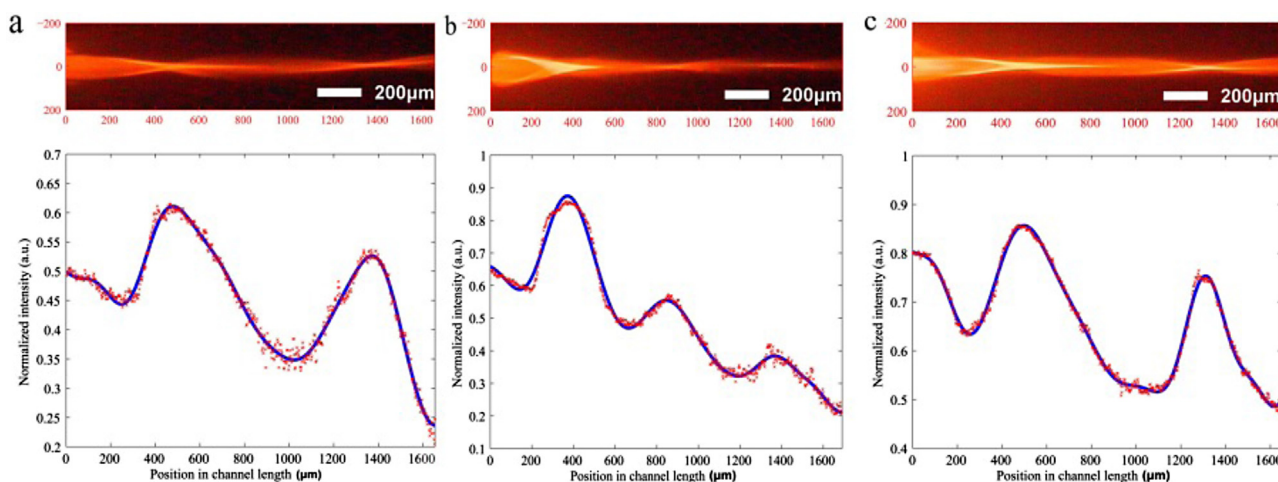


Fig. 3. Normalized fluorescence intensity profiles along the central line (*x* axis) of the waveguide at different mole fractions of EG ((a) 0.0698; (b) 0.242; (c) 0.4119). Red point, experiment results; Blue line, smoothed results. (For interpretation of the references to color in this text, the reader is referred to the web version of the article.)

Table 2

Volume fraction (x_1) of ethylene glycol (EG), mole fraction (x_2) of EG, dynamic viscosity and refractive index (n) of core liquid, experimental focus length (L_F) of the first section and binary mutual diffusion coefficient (D) of EG–water solution at different mole fraction of EG at 298.2 K and atmospheric pressure.

x_1	x_2	μ (mPa s)		n		L_F^a (μm)	D^a ($1\text{e}^{-10}\text{ m}^2/\text{s}$)	Literature value [11] ($1\text{e}^{-10}\text{ m}^2/\text{s}$)	
Core	Cladding	Core	Cladding	Core	Cladding				
20%	0	0.0323	1.46	0.90	1.3504	1.3325	593 ± 11.4	12.35 ± 0.56	10.67
40%	0	0.0698	2.57	0.90	1.3708	1.3325	480 ± 7.2	8.36 ± 0.29	9.72
60%	0	0.1140	4.52	0.90	1.3912	1.3325	435 ± 4.8	7.12 ± 0.19	8.76
80%	0	0.1667	8.29	0.90	1.4116	1.3325	404 ± 6.2	6.38 ± 0.21	7.75
100%	0	0.2420	15.90	0.90	1.4318	1.3325	384 ± 5.4	5.86 ± 0.16	6.58
100%	20%	0.3104	15.90	1.46	1.4318	1.3504	413 ± 3.9	5.02 ± 0.10	5.72
100%	40%	0.4119	15.90	2.57	1.4318	1.3708	498 ± 7.0	4.33 ± 0.13	4.75
100%	60%	0.5456	15.90	4.52	1.4318	1.3912	602 ± 9.6	3.20 ± 0.11	3.97
100%	80%	0.7298	15.90	8.29	1.4318	1.4116	778 ± 9.4	2.84 ± 0.09	3.44

Standard uncertainties u for temperature, volume fraction; dynamic viscosity, refractive index are $u(T)=0.1 \text{ K}$; $u(x_1)=0.01$; $u(\mu)=0.01 \text{ mPa s}$; $u(n)=0.0001$, respectively; Relative standard uncertainty u_r for focus length and mutual diffusion coefficient are $u_r(L_F)$ less than 1.9%; $u_r(D)$ less than 4.5%, respectively.

^a Standard deviations determined from three replicate experiments.

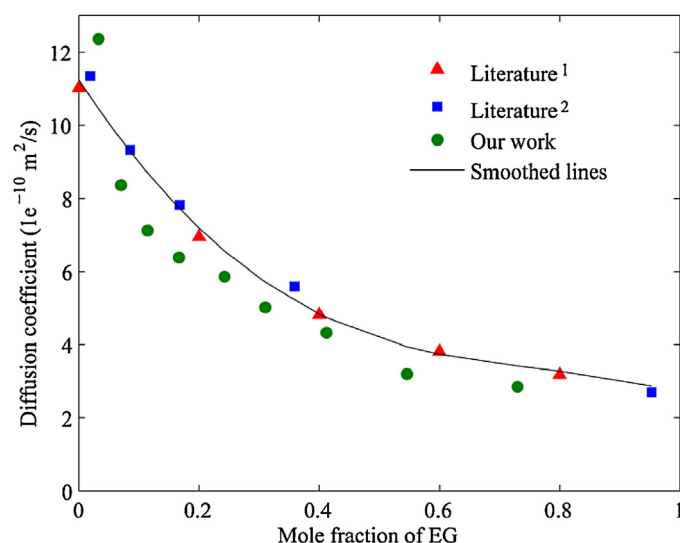


Fig. 4. Mutual diffusion coefficient of EG–water solution as a function of mole fraction of EG at 298.2 K and atmospheric pressure: ●, our work; ■, literature [11]; ▲, literature [26]; lines, smoothed lines with Refs. [11,26].

mutual diffusion coefficients of eight imidazolium-based ionic liquids ([Mmim][BF₄], [Emim][BF₄], [Bmim][BF₄], [Emim][N(CN)₂], [Bmim][N(CN)₂], [Emim][ClO₄], [Bmim][ClO₄] and [Bmim][NO₃]) in water were investigated. Fig. 5 shows normalized fluorescence

intensity profiles for three different RTILs ([Mmim][BF₄], [Emim][BF₄] and [Bmim][NO₃]) along the central line (x axis) of waveguide. The estimated mutual diffusion coefficients are listed in Table 3. There are altogether four anions BF₄[−], NO₃[−], ClO₄[−] and N(CN)₂[−] and three cations [Mmim]⁺, [Emim]⁺ and [Bmim]⁺ in the data set.

When we choose the same anion BF₄[−], the summation of the binary mutual diffusion coefficients, in order of decreasing values, are as follows: [Mmim][BF₄] > [Emim][BF₄] > [Bmim][BF₄], which indicates that the extension of the alkyl side-chain of the cation decreases mutual diffusion coefficient. Next, when we choose the same cation [Bmim]⁺, the summation of the binary mutual diffusion coefficients, in order of decreasing values, are as follows: [Bmim][NO₃] > [Bmim][N(CN)₂] > [Bmim][BF₄] > [Bmim][ClO₄]. It means that anion also affects the mutual diffusion coefficients.

It should be noted that we did not observe the focusing phenomenon for [Bmim][ClO₄] in the flow rates of $Q_{\text{cladding}} = 1 \mu\text{l/min}$ and $Q_{\text{core}} = 0.2 \mu\text{l/min}$, due to the mutual diffusion coefficient of [Bmim][ClO₄]–water being very small. To estimate the mutual diffusion coefficients of [Bmim][ClO₄]–water, we decreased the flow rates ($Q_{\text{cladding}} = 1 \mu\text{l/min}$ and $Q_{\text{core}} = 0.1 \mu\text{l/min}$) to make the core liquid and the two cladding liquids diffusive mixing as soon as possible in the waveguide. The focus length of the first section is $546 \pm 5.1 \mu\text{m}$, the mutual diffusion coefficient of [Bmim][ClO₄]–water is about $0.69 \pm 0.03 \times 10^{-10} \text{ m}^2/\text{s}$.

The summation of binary mutual diffusion coefficients of eight RTILs–water, in order of decreasing values, are as follows:

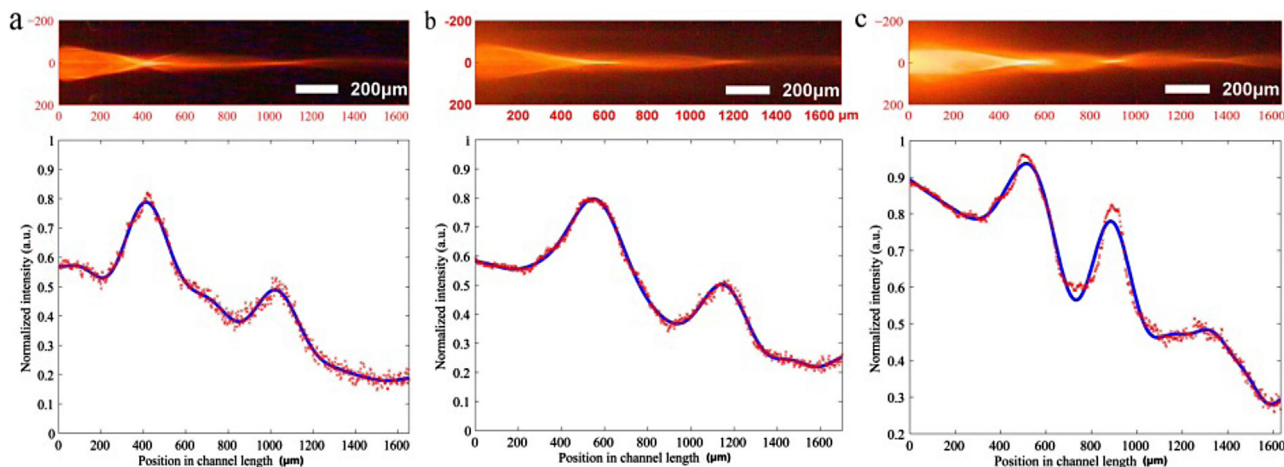


Fig. 5. Normalized fluorescence intensity profiles along the central line (x axis) of the waveguide for (a) [Mmim][BF₄], (b) [Emim][BF₄] and (c) [Bmim][NO₃]. Red point, experiment results; Blue line, smoothed results. (For interpretation of the references to color in this text, the reader is referred to the web version of the article.)

Table 3
Dynamic viscosity (μ), molecular weights and refractive index (n) of core liquid, experimental focus length (L_F) of the first section and mutual diffusion coefficients (D) of eight different ionic liquids–water binary systems at 298.2 K and atmospheric pressure.

RTILs	μ (mPa s)	Molecular weights	n	L_F (μm)	D ($10^{-10} \text{ m}^2/\text{s}$)
[Mmim][BF ₄]	34	184	1.4023	432 ± 6.4	6.87 ± 0.23
[Emim][BF ₄]	45	198	1.4098	548 ± 8.1	3.42 ± 0.12
[Bmim][BF ₄]	146	226	1.4184	889 ± 8.9	1.24 ± 0.05
[Emim][N(CN) ₂]	20	177	1.5127	348 ± 4.3	7.25 ± 0.18
[Bmim][N(CN) ₂]	32	205	1.5050	535 ± 6.5	2.53 ± 0.08
[Emim][ClO ₄]	68	211	1.4756	628 ± 8.2	1.93 ± 0.09
[Bmim][ClO ₄]	265	239	1.4736	–	–
[Bmim][NO ₃]	165	201	1.4926	526 ± 7.6	3.13 ± 0.10

– no focussing effect was observed at flow rate $Q_{\text{cladding}} = 1 \mu\text{l}/\text{min}$, $Q_{\text{core}} = 0.2 \mu\text{l}/\text{min}$.

Standard uncertainties u for temperature, dynamic viscosity, refractive index are $u(T) = 0.1 \text{ K}$; $u(\mu) = 0.5 \text{ mPa s}$; $u(n) = 0.0001$, respectively; relative standard uncertainty u_r for focus length and mutual diffusion coefficient $u_r(L_F)$ less than 1.5%; $u_r(D)$ less than 4.7%, respectively.

Table 4
Experimental focus length (L_F) of the first section and mutual diffusion coefficients (D) of three RTILs–water binary systems with the temperature range from 298.2 to 338.2 K at atmospheric pressure.

T (K)	[Mmim][BF ₄]		[Emim][BF ₄]		[Bmim][BF ₄]	
	L_F (μm)	D ($10^{-10} \text{ m}^2/\text{s}$)	L_F (μm)	D ($10^{-10} \text{ m}^2/\text{s}$)	L_F (μm)	D ($10^{-10} \text{ m}^2/\text{s}$)
298.2	432 ± 6.4	6.87 ± 0.23	548 ± 8.1	3.42 ± 0.12	889 ± 8.9	1.24 ± 0.05
308.2	411 ± 6.8	7.74 ± 0.33	475 ± 5.3	4.65 ± 0.14	643 ± 9.3	2.56 ± 0.10
318.2	398 ± 5.1	8.45 ± 0.27	443 ± 6.2	5.57 ± 0.25	508 ± 7.2	3.83 ± 0.16
328.2	377 ± 3.7	9.62 ± 0.29	408 ± 4.9	6.72 ± 0.23	454 ± 4.6	5.13 ± 0.17
338.2	354 ± 5.2	10.65 ± 0.38	374 ± 3.8	7.89 ± 0.20	413 ± 5.7	6.45 ± 0.28

Standard uncertainties u for temperature is $u(T) = 0.1 \text{ K}$; relative standard uncertainty u_r for focus length and mutual diffusion coefficient are $u_r(L_F)$ less than 1.5%; $u_r(D)$ less than 4.7%, respectively.

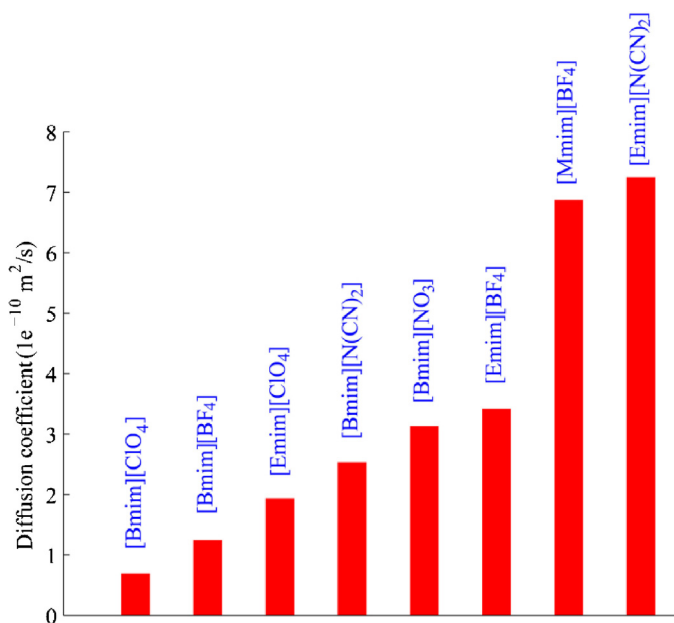


Fig. 6. Mutual diffusion coefficients of eight RTILs–water binary systems.

$[\text{Emim}][\text{N}(\text{CN})_2] > [\text{Mmim}][\text{BF}_4] > [\text{Emim}][\text{BF}_4] > [\text{Bmim}][\text{NO}_3] > [\text{Bmim}][\text{N}(\text{CN})_2] > [\text{Emim}][\text{ClO}_4] > [\text{Bmim}][\text{BF}_4] > [\text{Bmim}][\text{ClO}_4]$, as shown in Fig. 6. As is clear from these results, the value of mutual diffusion coefficient D is well correlated to molecular weight of the ionic liquids.

The mutual diffusion coefficients data of the [Mmim][BF₄], [Emim][BF₄] and [Bmim][BF₄] in water at temperatures ranging from 298.2 to 338.2 K are listed in Table 4. The refractive index and dynamic viscosity at different temperatures were measured (Table S.1). Fig. 7 shows the measurement values reported in Table 4. As would be expected, the diffusion coefficient increases quickly with temperature and is sensitive to it.

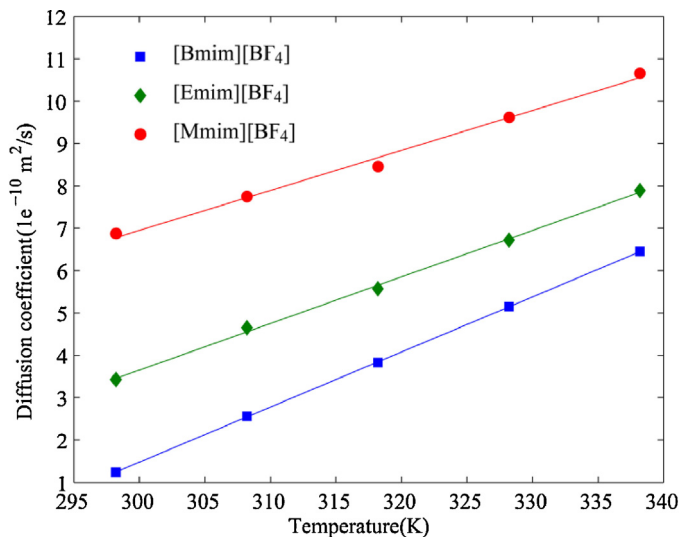


Fig. 7. Mutual diffusion coefficients of three ionic liquids–water binary systems as a function of temperature: (●), [Mmim][BF₄]; (◆), [Emim][BF₄]; (■), [Bmim][BF₄] this work; and lines, smoothed results.

4. Conclusion

Quantitative analysis of the mutual diffusion coefficient of ionic liquids–water is a necessity for the efficient design of microfluidic and optofluidic devices as well as an important physicochemical method in its own right. This study developed L^2 optofluidic waveguides as a research tool, employing the combination of experimental data with an analytical model to quantitatively describe molecular interactions. Mutual diffusion coefficients of aqueous solutions of EG at different volume fractions were measured to confirm the validity of our proposed method. Then, a series of mutual diffusion coefficients of RTILs–water were also estimated. Our results show that mutual diffusion coefficients are strongly affected by chemical composition (cations and anions) of the solute,

temperature and concentrations. A potential source of error (about 15%) in calculation of mutual diffusion coefficients with our method is the assumption of no molecule-wall interactions and PDMS microchannel wall is smooth perfectly in simulation. However, the molecule-wall interactions could no longer be neglected when the depth of the channel reaches a critical value. It is hoped that in future work, we will consider the more complex factors in simulation and fabricate smoother microchannel with new methods.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.fluid.2014.01.003>.

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